

Milk Distribution SaaS - Master PRD / BRD v3

Executive Summary

A SaaS platform connecting milk distributors and customers with subscriptions, delivery management, billing and administration.

Business Goals

Digitize dairy delivery operations, reduce manual records, automate billing, improve customer retention.

User Personas

Super Admin, Distributor, Customer.

Admin Features

Verification, platform monitoring, user management, subscriptions, deliveries, billing oversight, notifications.

Distributor Features

Customer onboarding, pricing, fat-based milk variants, delivery slots, subscriptions, billing, reports.

Customer Features

Distributor discovery, subscriptions, pause requests, extra milk requests, billing, notifications.

Product Catalog

Milk, Curd, Paneer, Eggs, Lassi, Ghee, Butter, Khoya.

Milk Variant System

Cow, Buffalo, Goat, Camel, custom variants with fat percentages and pricing.

Subscription Engine

Daily, Alternate Day, Weekdays, Weekly, Monthly. Future support for custom schedules.

Pause Logic

Customers can pause deliveries before configured cutoff time. Billing automatically adjusts.

Extra Milk Logic

Temporary quantity increases for specific dates.

Delivery Engine

Generate daily delivery lists from active subscriptions and adjustments.

Billing Engine

Calculate bills from deliveries, pricing and adjustments.

Notification Engine

Distributor-to-customer and platform-wide communications.

Distributor Approval Workflow

Register → Verification → Setup Wizard → Go Live.

Customer Journey

Register → Find Distributor → Subscribe → Receive Deliveries → Billing.

Database Domains

Users, Customers, Distributors, Pricing, Delivery Slots, Subscriptions, Deliveries, Bills, Payments, Notifications.

Permissions Matrix

Role-based access for Admin, Distributor and Customer.

API Domains

Auth, Distributor, Customer, Subscription, Billing, Notifications, Admin.

Security

JWT, refresh tokens, RBAC, audit logging, validation.

Testing Strategy

Unit, integration, E2E, UAT and production monitoring.

KPIs

Active subscriptions, revenue, delivery success rate, retention, distributor growth.

Future Roadmap

Payments, Wallets, Mobile Apps, Route Optimization, IoT integrations, Analytics, Loyalty Programs.

Phase-wise Roadmap

Phase 1: Core SaaS platform (current)

Phase 2: Payment gateway, wallet, OTP login, mobile optimization

Phase 3: Analytics, route optimization, loyalty system

Phase 4: IoT dairy ecosystem and smart automation